

# BAM as a Transformer add-on

BamLlama2MediumV2C256LocalFetchC8SharedReadLLFScan

24 layers / D = 1024 / 16 MHA heads × 64 / SwiGLU width = 2816

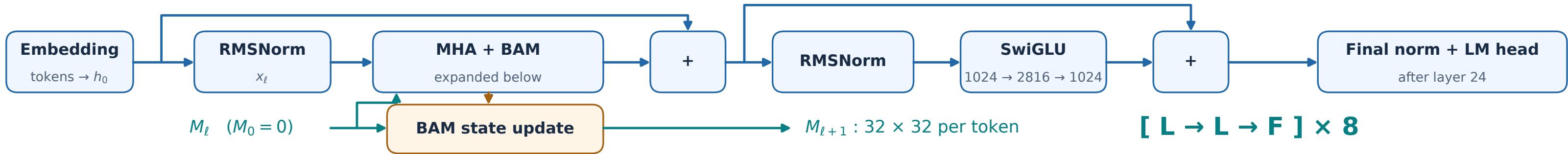
Transformer

BAM read / route

BAM write

## 01 / MODEL & DECODER

### Two streams through depth: residual vectors + per-token matrix state

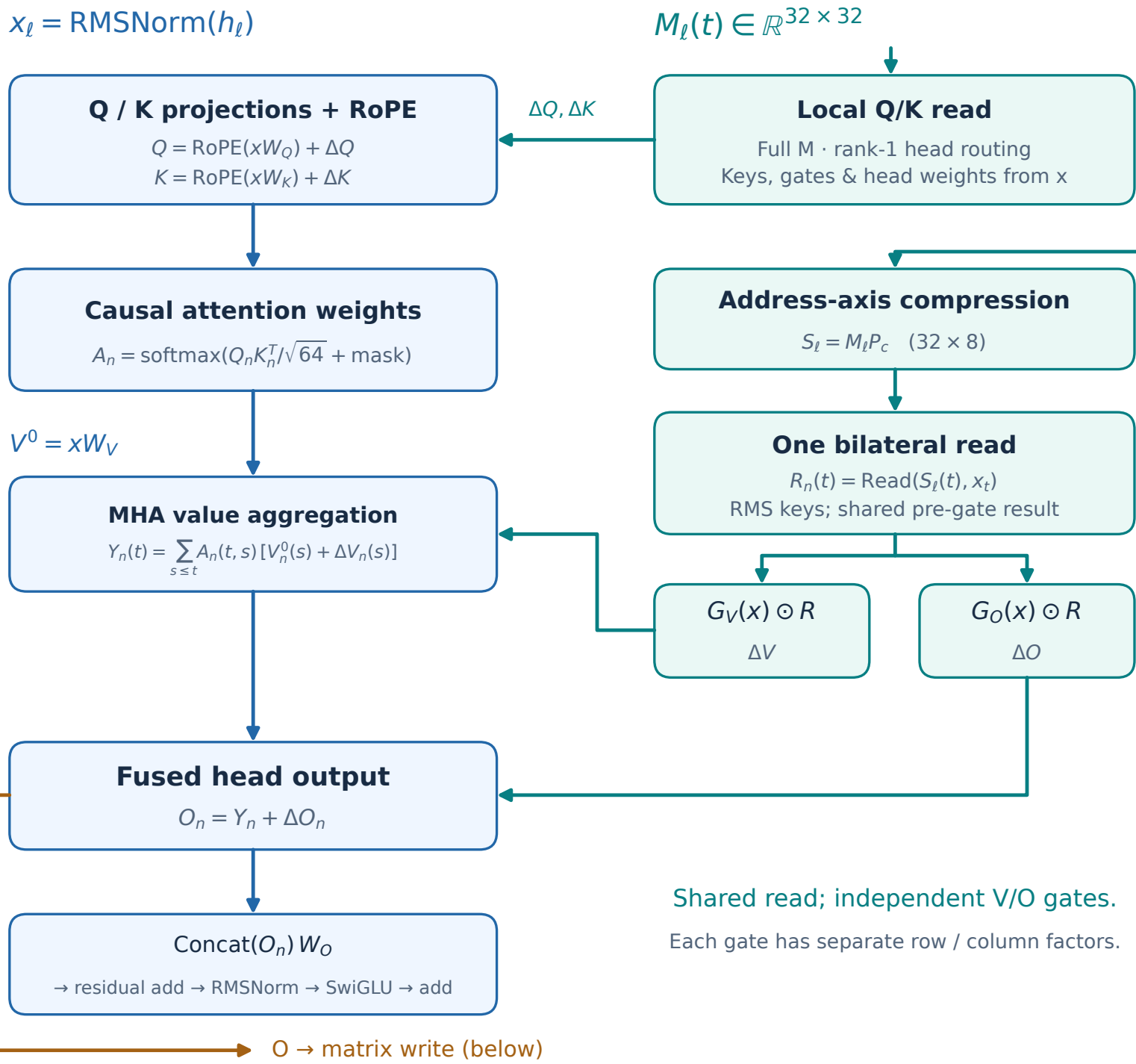


Repeat the decoder from RMSNorm to the second residual add; each layer has its own parameters.

## 02 / LOCAL LAYER

### L · Shared read into V and O

Layers 1, 2, 4, 5, ..., 22, 23



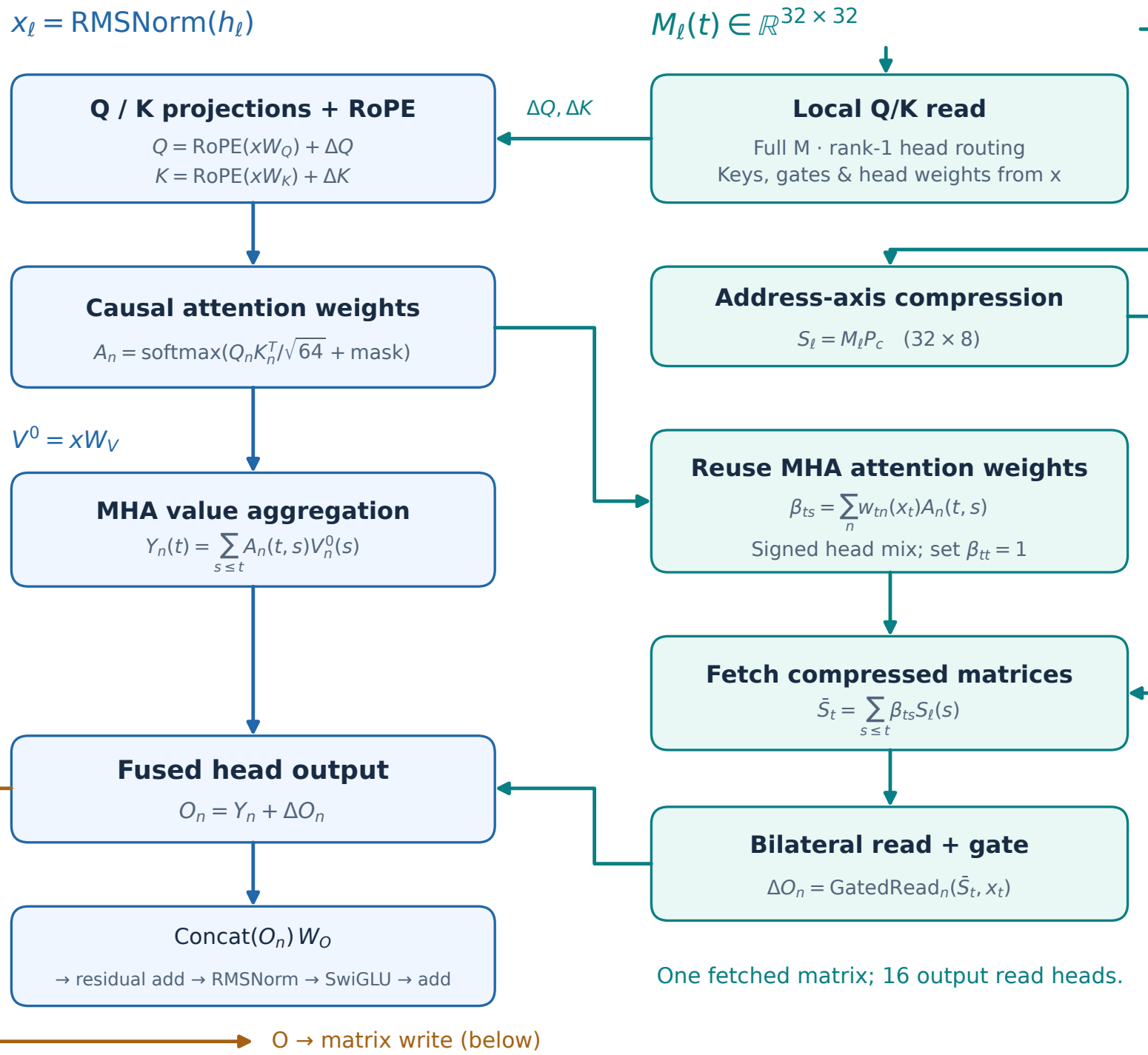
Shared read; independent V/O gates.

Each gate has separate row / column factors.

## 03 / FETCH LAYER

### F · MHA routes matrix retrieval

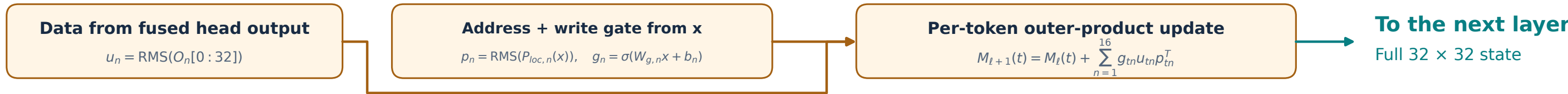
Layers 3, 6, 9, ..., 24



One fetched matrix; 16 output read heads.

## 04 / WRITE IN EVERY LAYER

### The fused MHA + BAM head output becomes new matrix content



Local = same-token BAM read. All 24 layers use global causal MHA. C256 = query tiling; C8 = compressed address width.

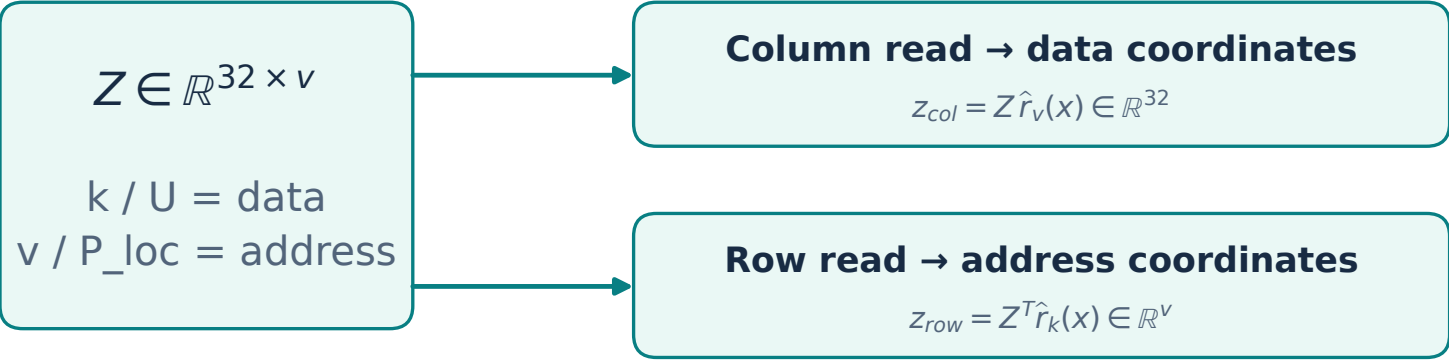
Layer scan executes [L,L,F] blocks with distinct layer parameters. Q/K additions are after RoPE; QK normalization is disabled.

# BAM read / write mechanics

Companion to the Transformer interaction diagram · same SharedRead LLF configuration

## A / BILATERAL READ

### One matrix, two directions



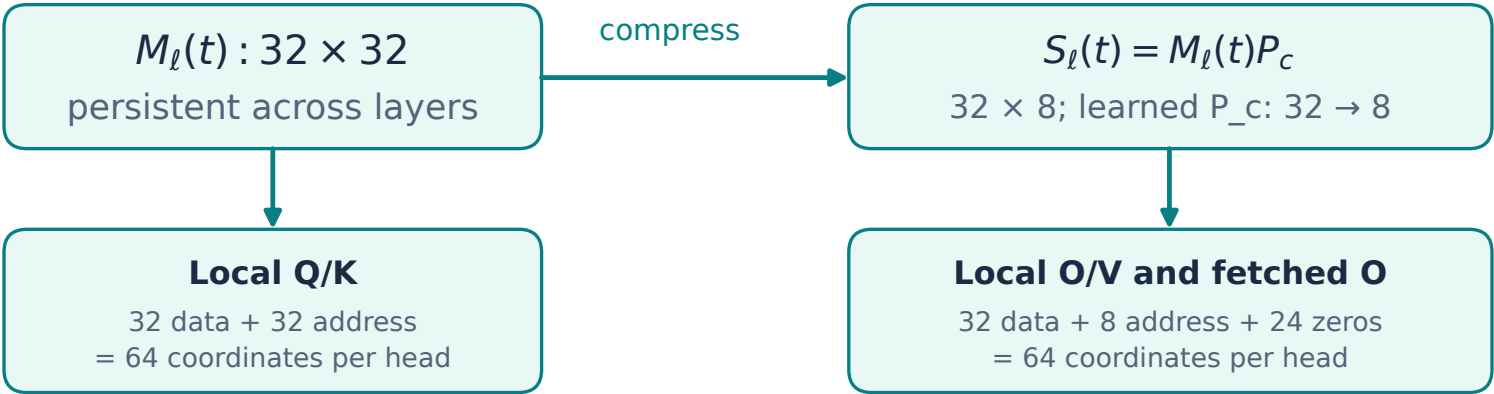
$\hat{r} = \text{RMS}(r)$  · row and column keys normalized separately

Read keys, routing weights and sigmoid gates are functions of x.

The persistent matrix stream has no whole-matrix RMS normalization.

## B / EXACT WIDTHS

### Full state and compressed read view



Compression changes only the read view; every write updates full M.

No learned 8 → 32 decoder: compressed row results occupy slots 32:40.

## C / LOCAL QK & SHARED O/V

### What is shared, and what is independent?

**Local Q/K:** one basis per read side, separately for Q and K.

$$\Delta Q_n = [a_{n,col}^Q z_{col}^Q ; a_{n,row}^Q z_{row}^Q]$$

Here z includes side gates; signed, RMS-normalized head weights route it.

K uses the same construction with its own keys, gates and head weights.

**Local O/V:** 16 read heads share one pre-gate result between destinations.

$$\Delta V_n = G_{V,n}(x) \odot R_n, \quad \Delta O_n = G_{O,n}(x) \odot R_n$$

$G_{V,n}, G_{O,n}$  each have row / column factors  $2\sigma(\cdot)$ . No cross-layer weight tying.

## D / FETCH ROUTE & WRITE

### Reuse attention for transport; write locally

$$w_t = \text{RMS}(W_{mix} x_t) / \sqrt{16}$$

$$\beta_{ts} = \sum_n w_{tn} A_n(t, s) \quad (s < t), \quad \beta_{tt} = 1$$

$$\bar{S}_t = S_\ell(t) + \sum_{s < t} \beta_{ts} S_\ell(s)$$

Signed route; causal / segment mask inherited from MHA; no second softmax.

$$P_{loc}(x) = W_{up} \text{GELU}(W_{down} x) + b$$

1024 → 256 → (16 × 32); one address factor per MHA head.

Write uses O before W\_O; retention = 1; data and address factors use RMS.

## E / READING THE TWO STREAMS

### M carries depth history; MHA supplies token-to-token communication

L layer:  $M(t) \rightarrow Q/K, V$  and O at token t; injected V is then transported by ordinary MHA.

F layer: the existing MHA map transports  $S(s)$  to t; a query-conditioned read converts the fetched matrix into O.

Both: fused O(t) supplies write data, while x(t) supplies the address and write gate; the FFN updates only the residual stream.